

ISEN/DAEN 427 Review for Exam #2

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- Prior + Likelihood $\rightarrow$ Posterior (Bayes' theorem)
- Prior informativeness depends on scale
- MCMC approximates posteriors when integrals are intractable

1. Beta Priors and Their Shapes

- **Beta(0.5, 0.5)**: U-shaped, favors extreme probabilities near 0 or 1.
- **Beta(1, 1)**: Uniform, non-informative prior.
- **Beta(1, 4)**: Right-skewed, favors small probabilities.

2. Components of a Probabilistic Model

- **Priors:** $\mu \sim N(0, 1)$, $\sigma \sim HN(1)$
- **Likelihood:** $Y|\mu, \sigma \sim N(\mu, \sigma)$
- **Posterior:** $p(\mu, \sigma|y) \propto p(y|\mu, \sigma)p(\mu)p(\sigma)$
- Posterior dimension equals number of parameters: 2 for (μ, σ) .

3. Bayes' Theorem Example: Loaded Coin

$$P(C_2|H) = \frac{P(H|C_2)P(C_2)}{P(H|C_1)P(C_1) + P(H|C_2)P(C_2)} = \frac{2}{3}$$

- Observing a head $\rightarrow$ 2/3 chance the coin was loaded.

4. Prior Informativeness

- Prior: $\sigma \sim HN(200 \text{ kg})$
- **Blue whales:** overly informative.
- **Humans:** weakly informative.
- **Mice:** non-informative.

Informativeness depends on the scale of the problem.

5. Response and Measurement Models

- Student's mistake: $p(\hat{s}|s) = p(x|s = \mu)p(s)$ is wrong.
- Correct: $p(\hat{s}|s) = \int p(\hat{s}|x)p(x|s)dx$
- More measurement noise $\rightarrow$ broader $p(\hat{s}|s)$.

6. Key Bayesian Concepts

- **Likelihood:** describes how data are generated given parameters.
- **Posterior:** combines prior and likelihood.
- **Evidence:** $p(D) = \int p(D|\theta)p(\theta)d\theta$ — often intractable.

7. MCMC Overview

- **Goal:** sample from $p(\theta|D)$ without computing $p(D)$.
- **Metropolis–Hastings:**
 - 1 Propose θ' from $q(\theta'|\theta_t)$.
 - 2 Compute $r = \frac{p(D|\theta')p(\theta')}{p(D|\theta_t)p(\theta_t)}$.
 - 3 Accept θ' with prob. $\alpha = \min(1, r)$.
- **Advantage:** flexible for complex posteriors.
- **Disadvantage:** slow convergence, correlated samples.

8. Gaussian–Gaussian Example

- Prior: $\theta \sim N(0, 4)$; Likelihood: $x = 5.2 \sim N(\theta, 1)$
- Posterior: $p(\theta|x) = N(4.16, 0.8)$
- MCMC acceptance for $\theta' = 3.0, \theta_t = 2.0$: $\alpha = 1$